MATTHEW C. PHARR

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EDUCATION

Columbia University, New York, NY
Ph.D. Applied Physics, 2026
M.Phil. Applied Physics, 2025
M.S. Applied Physics, 2023

Rensselaer Polytechnic Institute, Troy, NY
B.S. Physics & Mathematics, 2021
Summa Cum Laude

PUBLICATIONS

- **M. Pharr**, N. C. Logan, J. K. Park, and C. Paz-Soldan, *Resonant and non-resonant error field effects in drift-kinetically perturbed tokamak equilibria*, Manuscript in preparation (2026).
- **M. Pharr**, E. Bursch, N. C. Logan, P. Lunia, J. K. Park, and C. Paz-Soldan, *Coordinate-invariant flux-surface Fourier analysis in tokamaks*, Phys. Plasmas 33, 092503 (2026).
- **M. Pharr**, N. C. Logan, J. K. Park, and C. Paz-Soldan, *Quantifying resonant drive in resistive perturbed tokamak equilibria*, Nucl. Fusion 66, 086035 (2026).
- **M. Pharr**, N. C. Logan, C. Paz-Soldan, J. K. Park, and C. J. Hansen, *Error field predictability and consequences for ITER*, Nucl. Fusion 64, 126025 (2024).
- F. Ebrahimi and **M. Pharr**, *A nonlocal magneto-curvature instability in a differentially rotating disk*, Astrophys. J. 936, 145 (2022).

- N. C. Logan et al., *Unification and optimization of the Generalized Perturbed Equilibrium Code*, Manuscript in preparation (2026).
- S. Weber et al., *Fast and numerically robust error-field response from matrix Riccati integration*, Manuscript in preparation (2026).
- E. Bursch et al., *Improved $n = 1$ empirical error field penetration threshold scaling with Ohmic and L-mode conventional tokamak plasma discharges*, Submitted to Plasma Phys. Control. Fusion (2026).
- S. Freiberger et al., *Compact experimental negative triangularity reactor (CENTAUR): a design study for a compact, affordable breakeven tokamak*, Plasma Phys. Control. Fusion 68, 095025 (2026).
- N. C. Logan et al., *SPARC tokamak error field expectations and physics-based correction coil design*, Nucl. Fusion 66, 066003 (2026).
- X. Bai et al., *Time variation of error field correction in ITER*, Plasma Phys. Control. Fusion 67, 095013 (2025).
- S. Guizzo et al., *Electromagnetic system conceptual design for a negative triangularity tokamak*, Fusion Eng. Des. 219, 115257 (2025).
- The MANTA Collaboration et al., *MANTA: a negative-triangularity NASEM-compliant fusion pilot plant*, Plasma Phys. Control. Fusion 66, 105006 (2024).
- C. J. Hansen et al., *TokaMaker: an open-source time-dependent Grad-Shafranov tool for the design and modeling of axisymmetric fusion devices*, Comput. Phys. Commun. 298, 109111 (2024).
- C. Paz-Soldan et al., *Simultaneous access to high normalized density, current, pressure, and confinement in strongly-shaped diverted negative triangularity plasmas*, Nucl. Fusion 64, 094002 (2024).
- C. T. Holcomb et al., *DIII-D research to provide solutions for ITER and fusion energy*, Nucl. Fusion 64, 112003 (2024).

RESEARCH & PROJECTS

Monaco Postdoctoral Fellow, ITER Organization October 2026 – Present
Science Division
Supervisor: Dr. Simon Mcintosh.

- Apply linear MHD to set installation tolerances for PF coils.
- Develop a holistic resonant and nonresonant error field correction strategy for ITER.

Graduate Research Assistant, Columbia University August 2021 – August 2026
Department of Applied Physics and Applied Mathematics, New York, NY
Advisors: Dr. Carlos Paz-Soldan, Dr. Nikolas Logan.

- Conducted research in theoretical and computational plasma physics, focusing on 3D perturbations of tokamak equilibria, error field effects and their induced risks, and MHD stability.
- Created an error field source model for ITER, and used it to predict the risk of error field penetration in ITER's baseline scenario. Used this model to propose a novel method for ensuring extrapolability of error field measurements from low to high beta.
- Developed methods for quantifying the resonant drive of magnetic islands in non-ideal perturbed tokamak equilibria.
- Contributed to the MANTA project, a joint Columbia/MIT negative triangularity pilot plant design study. Worked on integrated modeling of plasma equilibria, transport, and 0D systems modeling.
- Developed and applied computational tools, including: GPEC, a fast linear perturbed equilibrium suite, including sub-modules DCON and its asymptotic matching code, RDCON/RMATCH, as well as its Julia-based modernization; TokaMaker, a flexible, open-source Grad-Shafranov solver for tokamak and dipole equilibria; OpenPOPCON, an open-source 0D tokamak systems code that has been used for scoping the MANTA and CENTAUR tokamak designs, as well as pedagogically and in consultation with fusion startups.
- Worked for the SPARC MHD team, including maintaining and developing its total error field source model, giving feedback on iterative coil design, and predicting time-dependent risk of error field penetration. Developed a module for the POPSIM code, a plasma control system simulator, to include effects of time-dependent error fields and their driven tearing modes and penetration risks.
- Attended MIT CPS-FR Computational Summer School, 2023.

SULI Intern, Princeton Plasma Physics Laboratory Fall 2020 – Summer 2021
Theory Department, Princeton, NJ
Supervisor: Dr. Fatima Ebrahimi.

- Studied large-scale magnetic field growth and stability in Hall-MHD simulations of quasi-Keplerian plasma flows, with applications to astrophysical accretion disks.
- Used the extended MHD code NIMROD to perform linear and nonlinear simulations of magnetized plasmas in cylindrical geometry.
- Discovered a new family of more global solutions to the flow-driven magnetorotational instability (MRI) through a hierarchy of models, from a new fast linear eigenvalue solver to full 3D nonlinear simulations on large-scale computing clusters.

Undergraduate Research Assistant, Rensselaer Polytechnic Institute Summer 2020 – Spring 2021
Department of Mathematical Sciences, Troy, NY
Supervisors: Dr. Peter Kramer, Dr. Scott Forth.

- Studied the application of stochastic methods to the modeling of complex biological systems, including meiotic spindle assembly.
- Used Python to implement a computational statistical model of microtubule dynamics, and performed data analysis and visualization.

PRESENTATIONS & CONFERENCE PROCEEDINGS

- **M. Pharr** et al., *Resistive and kinetic implications for ITER error field correction*, 47th ITPEA MDC Meeting, Seoul, Republic of Korea (2026).
- **M. Pharr** et al., *Resistive modifications to ideal error field sensitivity in tokamaks*, 30th Workshop on MHD Stability Control, San Diego, CA, USA (2026).
- **M. Pharr** et al., *Resistive and kinetic modifications to ideal error field sensitivity in tokamaks*, US-Japan MHD Workshop, Toki, Japan (2026).
- **M. Pharr** et al., *Quantifying the resonant drive for magnetic islands in perturbed ideal, resistive, and kinetic MHD equilibria*, 67th Annual Meeting of the APS Division of Plasma Physics, Long Beach, CA, USA (2025), abstract GO05.00007.

- **M. Pharr** et al., *Quantifying the resonant drive for magnetic islands in perturbed ideal, resistive, and kinetic MHD equilibria*, 29th US-Japan Joint Workshop on MHD Stability Control, Princeton, NJ, USA (2025).
- **M. Pharr** et al., *Expected error fields in ITER: a full-device source model and strategies for stable operation*, ITER International Summer School, Aix-en-Provence, France (2025).
- **M. Pharr** et al., *Expected error fields in ITER: a full-device source model and strategies for safe operation*, 66th Annual Meeting of the APS Division of Plasma Physics, Atlanta, GA, USA (2024), abstract UO06.00014.
- **M. Pharr** et al., *Error field predictability and consequences for ITER*, 65th Annual Meeting of the APS Division of Plasma Physics, Denver, CO, USA (2023), abstract PP11.00050.
- **M. Pharr** and F. Ebrahimi, *A nonlocal curvature-driven flow-shear instability in low-field plasmas*, Sherwood Fusion Theory Conference, Knoxville, TN, USA (2023).
- **M. Pharr** et al., *Error field source identification in early ITER plasmas*, 64th Annual Meeting of the APS Division of Plasma Physics, Spokane, WA, USA (2022), abstract PP11.00043.
- **M. Pharr**, F. Ebrahimi, and E. Blackman, *Large scale magnetic field growth and stability in Hall-MHD simulations of quasi-Keplerian flows*, 63rd Annual Meeting of the APS Division of Plasma Physics, Pittsburgh, PA, USA (2021), abstract BO06.00003.

HONORS & AWARDS

ORFEAS Fusion Design Contest Award; Columbia University 2022
For contributions to the Columbia/MIT team's project on negative triangularity reactor pilot plant design scoping. Awarded $22,000 as a group.

Max Hirsch Prize in Mathematics; Rensselaer Polytechnic Institute 2021
This prize is awarded to a Senior in the Department of Mathematical Sciences who has demonstrated outstanding ability in his or her academic work and also gives promise of outstanding success in a career in mathematical sciences.

J. Lawrence and Gertrude Katz Award in Physics; Rensselaer Polytechnic Institute 2021
This award is presented to the student selected as the outstanding graduating senior receiving a Bachelor of Science in Physics.

ΣΠΣ, Undergraduate Physics Honor Society 2021

Honorable Mention for Research Paper, Mathematical Contest in Modeling 2020

Rensselaer Archimedian Society (4.0/4.0 GPA award) 2019

Rensselaer Leadership Award 2018

PEDAGOGY

Adjunct Instructor, Intro Physics Lab Sequence Fall 2022 – Spring 2026
Barnard College of Columbia University, Physics and Astronomy, New York, NY

Senior Graduate Mentor, Fusion Reactor Design Fall 2024
Columbia University, Applied Physics and Applied Mathematics, New York, NY

Graduate Teaching Assistant, Complex Analysis/Linear Algebra Fall 2021 – Spring 2022
Columbia University, Applied Physics and Applied Mathematics, New York, NY

Undergraduate Facilitator Fall 2019 – Summer 2020, Spring 2021
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

I-PERSIST Mentor Fall 2019
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

ALAC Introductory Physics Tutor Fall 2019 – Spring 2020
Rensselaer Polytechnic Institute, Physics Department, Troy, NY

Private Tutor, Math/Physics Spring 2018 – Spring 2020
Self-employed. Provided 2–4 hours per week of university-level tutoring.

SKILLS

Languages and tools:
Python, Julia, Fortran 77/90/95, MATLAB, Linux, Git, LaTeX, OpenMP, MPI
Amazon Web Services, Oracle Cloud, Google Cloud
Fortran and C++ graphical / memory debugging tools
Linear/nonlinear MHD simulation and analysis with DCON, GPEC, NIMROD

General/practical skills:
High-performance computing
Analytic and numerical modeling of plasmas
Application of neural networks and machine learning to physics problems
Use of vacuum technology and other plasma-physics lab equipment
Use of high-voltage lab equipment
IT and computer networking
Native English, B1 French, introductory Italian, introductory Mandarin